📄 Product Requirements Document (PRD)

FEETFLOW – Intelligent Fleet Operations Platform

1. Product Overview:

Product Name: FEETFLOW
Category: Fleet Management SaaS Platform
Target Market: Delivery companies, logistics operators, courier services, last-mile delivery startups, SME fleet owners

🎯 Vision

To replace inefficient, paper-based fleet logbooks with a centralized, rule-based digital system that optimizes vehicle lifecycle, enhances driver safety, and maximizes financial performance.

🚨 Problem Statement

- Increased downtime
- Higher fuel & maintenance costs
- Unsafe driving practices
- Reduced profitability

2. 🎯 Objectives

FEETFLOW will:

- Digitize all fleet logbooks
- Automate maintenance scheduling
- Monitor driver behavior & safety
- Track operational & financial metrics
- Provide actionable analytics
- Improve compliance and accountability

3. 👥 Target Users

1️⃣ Fleet Owner / Admin

- Manages entire fleet
- Monitors expenses
- Sets rules & compliance policies

2️⃣ Operations Manager

- Assigns vehicles
- Monitors maintenance
- Tracks driver performance

3️⃣ Driver

- Logs trips digitally
- Receives alerts
- Submits incident reports

4️⃣ Accountant / Finance Manager

- Tracks cost per vehicle
- Generates profit reports

4. 🧩 Core Features

🚗 4.1 Fleet Lifecycle Management

Features:

- Vehicle registration database
- Digital vehicle logbook

- Service history records
- Insurance & permit tracking
- RC & document storage
- Automated maintenance reminders
- Downtime tracking
- Depreciation tracking

Rules Engine:

- Auto-alert if service overdue
- Alert if insurance expires within X days
- Notify if vehicle exceeds usage threshold

👮‍♀️ 4.2 Driver Safety & Monitoring

Features:

- Driver profile management
- License & document tracking
- Trip logging
- GPS-based trip tracking (Phase 2)
- Driving behavior score
- Over-speed alerts
- Harsh braking detection (API-based)
- Incident reporting system
- Driver performance scorecard

KPIs:

- Accident rate
- Average speed compliance
- Trip completion time
- Idle time percentage

💰 4.3 Financial Performance Tracking

Features:

- Fuel log entry
- Expense categorization
- Cost per km calculation
- Vehicle-wise revenue tracking
- Maintenance cost tracking
- Profit margin per vehicle
- Automated financial reports

Key Metrics:

- Cost per mile/km
- ROI per vehicle
- Fuel efficiency trends
- Monthly operational cost

📊 4.4 Smart Dashboard & Analytics

- Real-time fleet health overview
- Active vehicles vs inactive
- Revenue vs expenses graph
- Driver performance leaderboard
- Maintenance upcoming alerts
- Downloadable reports (PDF/Excel)

⚙️ 4.5 Rule-Based Automation Engine

Core differentiator of FEETFLOW.

Admins can configure rules like:

- IF fuel cost > X threshold → alert
- IF vehicle idle > Y hours → notify
- IF driver overspeed 3 times → penalty flag
- IF maintenance overdue → block vehicle assignment

5. Non-Functional Requirements

Requirement	Description
Security	Role-based access control
Scalability	Must support 10 → 10,000 vehicles
Availability	99% uptime
Performance	Dashboard load < 2 sec
Data Backup	Daily automated backup
Compliance	GDPR & local transport rules ready

6. 🏗️ System Architecture (High-Level)

Frontend:

- React / Next.js
- Tailwind CSS
- Chart.js / Recharts

Backend:

- Node.js / Express
- RESTful API
- JWT Authentication

Database:

- PostgreSQL (Relational Data)
- Redis (Caching)

Optional Phase 2:

- Telematics API integration
- IoT GPS device integration
- AI risk scoring engine

7. 🗄 Data Model (Core Entities)

- Users
- Vehicles
- Drivers
- Trips
- Fuel Logs
- Expenses
- Maintenance Records
- Rules
- Alerts
- Financial Reports

8. 🚀 MVP Scope (Hackathon Ready)

Must-Have:

- User Authentication
- Vehicle CRUD
- Driver CRUD

- Trip logging
- Fuel & expense entry
- Dashboard
- Basic rule alerts

Nice-to-Have:

- Report export
- Driver scoring
- Document upload

Not in MVP:

- IoT device integration
- AI prediction engine

9. 📈 Success Metrics

- Reduce manual log errors by 90%
- Reduce maintenance delays by 50%
- Improve fuel efficiency by 15%
- Reduce compliance penalties
- Improve driver safety score

10. 💼 Monetization Strategy

- SaaS Subscription Model
- Tiered Pricing:
  - Starter (Up to 10 vehicles)
  - Growth (Up to 100 vehicles)
  - Enterprise (Custom)

    Add-ons:
    - GPS integration

- AI risk module
- Custom analytics

11. 🔮 Future Roadmap

Phase 2:

- AI-powered predictive maintenance
- Route optimization
- Driver fatigue detection
- Insurance integration
- Mobile app (Driver version)

Phase 3:

- Blockchain-based compliance logs
- Carbon footprint tracking
- EV fleet management

💧 FEETFLOW Positioning Statement

FEETFLOW is an intelligent, rule-driven fleet lifecycle platform that transforms delivery operations from paper-based chaos into data-driven efficiency.